CURIONEST

Acids & Bases: Theories and Strength

Complete Lesson Pack · Brønsted–Lowry · pH & pOH · Acid Strength

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

Learning Objectives

1. Identify Brønsted–Lowry acids, bases, and conjugate acid–base pairs
2. Write equations for acid and base ionization reactions
3. Use the ion-product constant for water (Kw) to find $[H_3O^+]$ and $[OH^-]$
4. Convert between pH, pOH, $[H_3O^+]$, and $[OH^-]$ at 25 °C
5. Compare acid strength using Ka and percent ionization
6. Describe the acid–base behavior of amphiprotic species

Concept 1 — Defining Acids & Bases (Brønsted–Lowry)

An acid is any species that donates a proton (H^+) to another species; a base is any species that accepts a proton. This is the Brønsted–Lowry definition, and it is more general than the Arrhenius definition (which requires the production of H_3O^+ or OH^- in water).

In any Brønsted–Lowry reaction, two conjugate pairs are present: when an acid donates a proton, the remaining species is its conjugate base; when a base accepts a proton, the product is its conjugate acid.

Example: In the reaction $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$, water donates a proton to ammonia. Water is the acid and its conjugate base is OH^-; ammonia is the base and its conjugate acid is NH_4^+.

Amphiprotic species (such as H_2O, HCO_3^-, and HSO_4^-) can either donate or accept a proton depending on the other reactant.

★ Worked Example

Label the acid, base, conjugate acid, and conjugate base in the reaction: $HCN(aq) + H_2O(l) \rightleftharpoons CN^-(aq) + H_3O^+(aq)$

Solution: HCN donates a proton to water: HCN is the acid and CN^- is its conjugate base. H_2O accepts the proton: water is the base and H_3O^+ is its conjugate acid.

★ **Common mistake:** Confusing 'conjugate' with 'concentration' — a conjugate pair differs by exactly one proton (H^+).

★ **Common mistake:** Forgetting that H_3O^+ (hydronium) is the form H^+ takes in water.

Concept 2 — Water Autoionization and the Ion-Product Constant (Kw)

Water autoionizes to a tiny extent: $H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$.

At 25 °C, the ion-product constant is $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$.

In pure water, $[H_3O^+] = [OH^-] = 1.0 \times 10^{-7}$ M. Kw increases with temperature because autoionization is endothermic ($K_w \approx 5.6 \times 10^{-13}$ at 100 °C).

★ Worked Example

At 25 °C, a solution has $[OH^-] = 2.5 \times 10^{-4}$ M. Find $[H_3O^+]$ and decide if the solution is acidic, basic, or neutral.

Solution: $[H_3O^+] = K_w / [OH^-] = (1.0 \times 10^{-14}) / (2.5 \times 10^{-4}) = 4.0 \times 10^{-11}$ M. Since $[H_3O^+] < [OH^-]$, the solution is basic.

★ **Common mistake:** Using $K_w = 1.0 \times 10^{-14}$ at temperatures other than 25 °C.

★ **Common mistake:** Multiplying instead of dividing when solving for a missing ion concentration.

Concept 3 — pH, pOH, and the pH Scale

$pH = -\log[H_3O^+]$ and $pOH = -\log[OH^-]$. At 25 °C, pH + pOH = 14.00 for any aqueous solution.

Acidic solutions: $[H_3O^+] > 1.0 \times 10^{-7}$ M and pH < 7. Basic: $[H_3O^+] < 1.0 \times 10^{-7}$ M and pH > 7. Neutral: pH = 7 at 25 °C.

A one-unit pH change corresponds to a tenfold change in $[H_3O^+]$.

★ Worked Example

What are the pH and pOH of a 0.010 M solution of HCl (a strong acid that ionizes completely)?

Solution: $[H_3O^+] = 0.010$ M, so $pH = -\log(0.010) = 2.00$. $pOH = 14.00 - 2.00 = 12.00$.

★ **Common mistake:** Forgetting the minus sign: $pH = -\log[H_3O^+]$, not $\log[H_3O^+]$.

★ **Common mistake:** Assuming pH = 7 is neutral at all temperatures (it is only true at 25 °C).

Concept 4 — Relative Strengths of Acids and Bases

Strong acids (HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$) ionize completely in water. Weak acids (HF, CH_3COOH, HCN, HNO_2) ionize only partially.

The strength of a weak acid is quantified by its ionization constant Ka: the larger Ka, the stronger the acid. For a conjugate pair, $K_a \times K_b = K_w$.

Water has a leveling effect: in water, the strongest acid possible is H_3O^+ and the strongest base possible is OH^-. The stronger the acid, the weaker its conjugate base.

Binary acid strength increases left to right across a period (CH_4 < NH_3 < H_2O < HF) and down a group (HF < HCl < HBr < HI).

★ Worked Example

For HF ($K_a = 6.8 \times 10^{-4}$) and HCN ($K_a = 6.2 \times 10^{-10}$), which is the stronger acid, and which is the stronger conjugate base?

Solution: HF has the larger Ka, so HF is the stronger acid and CN^- is the stronger conjugate base (the weaker acid has the stronger conjugate base).

★ Common mistake: Thinking 'strong' means 'concentrated' — strength depends on Ka (extent of ionization), not concentration.

★ Common mistake: Assuming all acids with H in the formula are strong.

Part A — Brønsted–Lowry Concepts (Questions 1–8)

1. Write an equation showing water acting as a Brønsted–Lowry acid when it reacts with ammonia, NH_3.

2. Write an equation showing water acting as a Brønsted–Lowry base when it reacts with hydrogen cyanide, HCN.

3. For each species, write the formula of its conjugate base: (a) H_2O (b) NH_4^+ (c) HSO_4^- (d) H_2CO_3

4. For each species, write the formula of its conjugate acid: (a) OH^- (b) NH_3 (c) CO_3^{2-} (d) $H_2PO_4^-$

5. In the reaction $HSO_4^-(aq) + H_2O(l) \rightleftharpoons SO_4^{2-}(aq) + H_3O^+(aq)$, identify the acid, base, conjugate acid, and conjugate base.

6. Which of the following species are amphiprotic? H_2O, OH^-, HCO_3^-, NH_4^+, HSO_4^-
   (A) H_2O and HSO_4^- only
   (B) H_2O, HCO_3^-, and HSO_4^-
   (C) OH^- and NH_4^+
   (D) All five species

7. Write the two possible reactions of the amphiprotic ion HCO_3^-: (a) as an acid with water (b) as a base with water.

8. Is the autoionization of water endothermic or exothermic? Explain using the values $K_w = 2.9 \times 10^{-14}$ at 40 °C and $K_w = 9.3 \times 10^{-14}$ at 60 °C.

Part B — Kw, pH, and pOH Calculations (Questions 9–16)

9. Calculate $[H_3O^+]$ and $[OH^-]$ in pure water at 25 °C.

10. A solution at 25 °C has $[H_3O^+] = 3.2 \times 10^{-5}$ M. Calculate $[OH^-]$. Is the solution acidic, basic, or neutral?

11. Calculate the pH and pOH of a 0.035 M solution of HNO_3, which ionizes completely.

12. Calculate the pH and pOH of a 0.0042 M solution of KOH, which ionizes completely.

13. What are [H3O+] and [OH−] in a solution whose pH is 4.20?

14. The [H3O+] in a sample of rainwater is 1.7 × 10−6 M at 25 °C. What is the concentration of hydroxide ions?

15. The [OH−] in household ammonia is 3.2 × 10−3 M at 25 °C. What is [H3O+] and the pH?

16. A solution has pH = 9.35. Calculate pOH, [H3O+], and [OH−]. Classify the solution.

Part C — Acid Strength (Questions 17–22)

17. Arrange the following in order of increasing acid strength: HF, HCl, HBr, HI. Explain the trend.

18. Given Ka values: CH3COOH (1.8 × 10−5), HF (6.8 × 10−4), HCN (6.2 × 10−10). Rank the acids from weakest to strongest.

19. For the conjugate pair HCN/CN−, Ka(HCN) = 6.2 × 10−10. Calculate Kb for CN−.

20. A 0.10 M solution of a weak acid HA is 2.4% ionized. Calculate [H3O+] and Ka.

21. The pH of a 0.23 M solution of HF is 1.92. Determine Ka for HF.

22. Explain why 'strong acid' does not mean 'concentrated acid.' Give an example of a dilute solution of a strong acid and a concentrated solution of a weak acid.

Key Terms

Term	Definition
acid	A species that donates a proton (H+) in a Brønsted–Lowry reaction
base	A species that accepts a proton (H+) in a Brønsted–Lowry reaction
conjugate acid	The species formed when a base accepts a proton

conjugate base	The species remaining after an acid donates a proton
amphiprotic	Able to act as either a proton donor or a proton acceptor
autoionization	Reaction of identical molecules to produce ions (e.g., water producing H_3O^+ and OH^-)
ion-product constant (K_w)	$K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C
pH	$-\log[H_3O^+]$, a measure of hydronium ion concentration
pOH	$-\log[OH^-]$, a measure of hydroxide ion concentration
ionization constant (K_a)	Equilibrium constant for the ionization of a weak acid; larger K_a = stronger acid

Answer Key

Worked solutions for every question — see answers embedded in Parts A–C above. The full key with step-by-step work is included in the teacher edition.

Part A — Brønsted–Lowry Concepts (Questions 1–8)

1. $H_2O(l) + NH_3(aq) \rightleftharpoons OH^-(aq) + NH_4^+(aq)$

2. $HCN(aq) + H_2O(l) \rightleftharpoons CN^-(aq) + H_3O^+(aq)$

3. (a) OH^- (b) NH_3 (c) SO_4^{2-} (d) HCO_3^-

4. (a) H_2O (b) NH_4^+ (c) HCO_3^- (d) H_3PO_4

5. Acid: HSO_4^- · Base: H_2O · Conjugate acid: H_3O^+ · Conjugate base: SO_4^{2-}

6. H_2O, HCO_3^-, and HSO_4^- (each can donate OR accept a proton; OH^- can only accept and NH_4^+ can only donate)

Why: Amphiprotic species must have at least one H to donate and be able to accept another H^+. OH^- has no H to donate; NH_4^+ has no lone pair to accept a proton easily in water.

7. (a) $HCO_3^-(aq) + H_2O(l) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$ (b) $HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq) + OH^-(aq)$

8. Endothermic — K_w increases with temperature, so heat is absorbed (an endothermic process shifts toward products when temperature rises).

Part B — Kw, pH, and pOH Calculations (Questions 9–16)

9. $[H_3O^+] = [OH^-] = 1.0 \times 10^{-7}$ M ($K_w = 1.0 \times 10^{-14}$; $x^2 = K_w \rightarrow x = 1.0 \times 10^{-7}$)

10. $[OH^-] = K_w/[H_3O^+] = 1.0\times10^{-14} / 3.2\times10^{-5} = 3.1 \times 10^{-10}$ M. $[H_3O^+] > [OH^-]$, so acidic.

11. $pH = -\log(0.035) = 1.46$; $pOH = 14.00 - 1.46 = 12.54$

12. $pOH = -\log(0.0042) = 2.38$; $pH = 14.00 - 2.38 = 11.62$

13. $[H_3O^+] = 10^{-4.20} = 6.3 \times 10^{-5}$ M; $[OH^-] = K_w/[H_3O^+] = 1.0\times10^{-14} / 6.3\times10^{-5} = 1.6 \times 10^{-10}$ M

14. $[OH^-] = 1.0\times10^{-14} / 1.7\times10^{-6} = 5.9 \times 10^{-9}$ M

15. $[H_3O^+] = 1.0\times10^{-14} / 3.2\times10^{-3} = 3.1 \times 10^{-12}$ M; $pH = -\log(3.1\times10^{-12}) = 11.51$

16. $pOH = 4.65$; $[H_3O^+] = 10^{-9.35} = 4.5 \times 10^{-10}$ M; $[OH^-] = 1.0\times10^{-14}/4.5\times10^{-10} = 2.2 \times 10^{-5}$ M. $pH > 7 \rightarrow$ basic.

Part C — Acid Strength (Questions 17–22)

17. HF < HCl < HBr < HI. Down group 17, the H–X bond weakens (bond enthalpy decreases), so ionization is easier.

18. $HCN < CH_3COOH < HF$ (smaller K_a = weaker acid)

19. $K_b = K_w/K_a = 1.0\times10^{-14} / 6.2\times10^{-10} = 1.6 \times 10^{-5}$

20. $[H_3O^+] = 0.10 \times 0.024 = 2.4 \times 10^{-3}$ M. $K_a \approx (2.4\times10^{-3})^2 / 0.10 = 5.8 \times 10^{-5}$ (using $[HA] \approx 0.10$ M, valid because % ionization < 5%).

21. $[H_3O^+] = 10^{-1.92} = 1.2 \times 10^{-2}$ M. $K_a = (1.2\times10^{-2})^2/(0.23 - 1.2\times10^{-2}) = 6.6 \times 10^{-4}$

22. Strength = extent of ionization (K_a); concentration = amount of solute per liter. A 0.0001 M HCl (strong, dilute) has lower $[H_3O^+]$ than 6 M CH_3COOH (weak, concentrated) even though HCl ionizes completely.